Day 099 | Unit 07 | Teacher plan

Understand and Use Percent

Check the Reference Whole

Standards: 6.RP.A.3c

Learning target: I can find the whole when a part and percent are known.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

Worked example 1

50% of a number is 60. Find the number and check.

1. Write part = decimal rate × whole.

2. Divide the part by the decimal rate to find the whole.

3. Whole = 60 ÷ 1/2 = 120.

4. Check 50% × 120 = 60.

5. State the result in the form requested: 120.

Answer: 120

Worked example 2

18 is 15% of an unknown quantity w. Select all equations that find w.
A. (15/100)w = 18
B. w = 18 × (15/100)
C. w = 18 + 15
D. w = 18 ÷ (15/100)

1. Write part = decimal rate × whole.

2. The part is percent rate times whole.

3. Divide 18 by 3/20

4. w = 120.

5. Check by multiplication.

Answer: A, D - (15/100)w = 18; w = 18 ÷ (15/100)

Day 099 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

20% of a number is 36. Find the number and check.

02 / Constructed response

50% of a number is 30. Find the number and check.

03 / Constructed response

50% of a number is 50. Find the number and check.

04 / Constructed response

12% of a number is 144/5. Find the number and check.

05 / Constructed response

25% of a number is 30. Find the number and check.

06 / Constructed response

125% of a number is 200. Find the number and check.

07 / Constructed response

48 students are 30% of a school’s enrollment. Find the enrollment.

08 / Multiple select

24 is 60% of an unknown quantity w. Select all equations that find w.
A. w = 24 × (60/100)
B. w = 24 + 60
C. w = 24 ÷ (60/100)
D. (60/100)w = 24

09 / Multiple select

12 is 30% of an unknown quantity w. Select all equations that find w.
A. (30/100)w = 12
B. w = 12 ÷ (30/100)
C. w = 12 × (30/100)
D. w = 12 + 30

10 / Constructed response

80 students are 40% of a school’s enrollment. Find the enrollment.

Day 099 | Unit 07 | Answer key

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1. 180

Whole = 36 ÷ 1/5 = 180. Check 20% × 180 = 36.

6.RP.A.3c | Constructed response

2. 60

Whole = 30 ÷ 1/2 = 60. Check 50% × 60 = 30.

6.RP.A.3c | Constructed response

3. 100

Whole = 50 ÷ 1/2 = 100. Check 50% × 100 = 50.

6.RP.A.3c | Constructed response

4. 240

Whole = 144/5 ÷ 3/25 = 240. Check 12% × 240 = 144/5.

6.RP.A.3c | Constructed response

5. 120

Whole = 30 ÷ 1/4 = 120. Check 25% × 120 = 30.

6.RP.A.3c | Constructed response

6. 160

Whole = 200 ÷ 5/4 = 160. Check 125% × 160 = 200.

6.RP.A.3c | Constructed response

7. 160

Whole = 48 ÷ 3/10 = 160. Check 30% × 160 = 48.

6.RP.A.3c | Constructed response

8. C, D - w = 24 ÷ (60/100); (60/100)w = 24

The part is percent rate times whole. Divide 24 by 3/5; w = 40. Check by multiplication.

6.RP.A.3c | Multiple select

9. A, B - (30/100)w = 12; w = 12 ÷ (30/100)

The part is percent rate times whole. Divide 12 by 3/10; w = 40. Check by multiplication.

6.RP.A.3c | Multiple select

10. 200

Whole = 80 ÷ 2/5 = 200. Check 40% × 200 = 80.

6.RP.A.3c | Constructed response